Derek Wang

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Education

University of Waterloo — **Waterloo, ON, Canada**
Bachelor of Applied Science in Computer Engineering — *Expected Apr 2030*

- Relevant Coursework: Fundamentals of Programming, Linear Algebra for Engineering, Project Studio

Experience

Autonomy Developer — **Aug 2025 – Present**
Waterloo Aerial Robotics Group — *Waterloo, ON, Canada*

- Developed Hardware-in-the-Loop (HITL) simulation systems in Python to address autonomous flight testing requirements for the 2026 National Annual UAS Student Competition
- Engineered image and GPS synchronization pipeline with panning capabilities to solve real-time sensor data correlation
- Redesigned GPS movement architecture using periodic functions to eliminate unrealistic teleportation behavior, creating smooth trajectory interpolation systems that enhanced simulation realism and accuracy by 75%

Robotics Software Programmer – VEX Robotics Competition — **Sept 2024 – May 2025**
Checkmate Robotics Club — *Markham, Ontario*

- Programmed competition software for Team 16868C Rushdown in the VEX Robotics Competition, securing invitation to the 2025 VRC World Championships among top 10% of global teams
- Deployed advanced robotics systems using Okapi PROS Library, Odometry positioning, and PID Controllers, boosting robot performance by 25% through precise motion control and comprehensive error handling
- Designed and refactored autonomous navigation algorithms, creating 15-second match routes and 60-second programming skills routines that achieved 5th place out of 80 teams at Provincial Championships

Robot Designer and Builder – FIRST Tech Challenge — **July 2019 – May 2022**
Dr. X Robotics Academy — *Richmond Hill, Ontario*

- Architected hardware solutions for Team 16417 Dr. X Academy in the FIRST Tech Challenge, earning invitation to the 2022 FTC World Championships among top 8% of global teams
- Designed and built 8 iterations of the team's robot, developing specialized components including drivetrain, arm, claw, turret systems and a backup robot for competition reliability
- Implemented autonomous control systems using Java, creating 30-second match routines with Dead Reckoning position estimation for precise navigation

Projects

Autonomous Vehicle Control System | C++ | Docker | CMake | Foxglove | *Source Code* | *Demo Video*

- Engineered LiDAR-based perception pipeline processing sensor data to generate real-time costmaps with 0.1m resolution, integrating persistent memory mapping system for 95% accurate dynamic environment representation
- Orchestrated autonomous navigation stack featuring A* path planning algorithm and Pure Pursuit tracking algorithm, achieving sub-0.5m waypoint accuracy and 100% obstacle avoidance success rate across 50+ test scenarios

Fruit Ninja AI | Python | YOLOv11 | *Source Code*

- Architected and implemented an automated computer vision pipeline using YOLOv11 object detection, achieving 90% accuracy in fruit classification
- Built comprehensive dataset of 816 images (514 training, 302 testing), optimizing model performance through strategic data augmentation and validation techniques

TTC Escape Puzzle Game | C# | Unity | Git | *Source Code*

- Designed innovative puzzle game concept for the 2025 GMTK Game Jam, leading a team of 3 developers
- Developed and implemented core gameplay mechanics using Unity and C#, creating a Toronto Transit Commission-inspired puzzle game featuring looping mechanics that enhanced player engagement

Skills

Languages: Java, C, C++, Python, HTML5, CSS, JavaScript, C#, TypeScript
Engineering Tools: Git, Fusion 360, Docker, Foxglove, Unity
Technologies: YOLOv11, ReactJS, OpenCV, Tailwind CSS